

Fractional Wiener chaos: Part 2

Interface spectral chaos and decay of collective correlations

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Abstract

We construct a product spectral decomposition from power-normalised parabolic-cylinder functions. At noninteger real orders, these functions are not square integrable with respect to the corresponding full-line Gaussian measure. Bringing the admissible half-line branches into the same Gaussian space and matching them at the origin gives a corrected orthonormal eigenbasis. A unitary transformation to a weighted probability space makes the eigenfunction corresponding to the lowest eigenvalue constant and permits consistent countable products. The resulting law is non-Gaussian under deformation, and the decomposition recovers classical Wiener chaos when the deformation is removed. We identify the associated closed gradient, adjoint divergence and deformed number operator.

The Rodrigues formula covers all orders occurring in the spectral branches, including negative orders. Under deformation, however, changing both branch orders by the same nonzero amount breaks the required relation between them. Fractional evolution defined through spectral calculus preserves the basis. After unitary identification, the one-coordinate fractional evolution operators converge in operator norm to Gaussian and three-dimensional radial Ornstein-Uhlenbeck fractional evolutions at the respective parameter endpoints. Convergence is uniform on compact time intervals away from zero.

We use the constructed product spectral decomposition to derive explicit expansions for collective threshold correlations under shared and independent inverse stable clocks. These expansions determine the algebraic decay rates and leading coefficients and provide explicit truncation error bounds. Under equilibrium initialisation, both models have identical one-coordinate process laws and identical fixed-time joint laws, yet collective correlations decay more slowly under a shared clock.

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1 Introduction

Classical Wiener chaos connects Hermite polynomials, multiple stochastic integrals and the eigenspaces of the Ornstein–Uhlenbeck operator [8, 12, 13]. The same integer is a Hermite order, an integration order and a spectral label. A fractional extension asks which of these connections survive when the Hermite order is allowed to be noninteger.

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This paper develops the spectral part of that programme. Its main construction starts with a corrected one-coordinate basis, changes the measure so that the positive eigenfunction corresponding to the lowest eigenvalue is constant, and takes products of the resulting eigenfunctions. We call this positive eigenfunction the first eigenfunction. This gives an orthogonal decomposition with a closed gradient and a diagonal number operator. We use the decomposition to compute how a collective threshold observation remains correlated with its initial value under fractional time evolution.

Part I [7] used power-normalised parabolic-cylinder functions, which coincide with the probabilists' Hermite polynomials at nonnegative integer orders. At other real orders their growth on the negative half-line prevents Gaussian square integrability. The proposed joining procedure also used branches normalised under different Gaussian measures. We correct the construction by transporting both branches into the same Hilbert space, specifying a common operator, and matching function values and first derivatives at the origin. The matching conditions select discrete admissible orders from the continuous special-function family.

Removing the Gaussian weight identifies this operator with an asymmetric harmonic oscillator. Its parabolic-cylinder branches and matching formulas are established in the oscillator literature [9]; the affine order relation and characteristic equation also appear in Chadzitaskos and Patera (arXiv:2204.06415, Eqs. (8), (11)). We use this operator to give the corrected Gaussian normalisation, domain and completeness argument required here.

The next step addresses the product construction. An unused coordinate must contribute the constant factor one if a finite-coordinate function is to remain unchanged when more coordinates are added. The deformed first eigenfunction is positive but nonconstant. Dividing every eigenfunction by it, and weighting the Gaussian law by its square, makes the first eigenfunction one and preserves orthonormality. This is the standard ground-state transformation [11, Chapter 1]. The transformed operator generates a conservative diffusion, and sums of its coordinate eigenvalues determine the product decomposition.

The contribution is the resulting spectral framework for this interface family, together with its domains, parameter limits and computable threshold coefficients. The tensor-product and ground-state transform methods are standard, as are general non-Gaussian chaos representations [23] and differentiable-measure calculi [6, 10]. At the undeformed endpoint our construction recovers Gaussian chaos and its gradient-divergence calculus [18]. Under deformation the product law is singular with respect to the Gaussian product law on the same coordinates. The term *spectral chaos* here means the orthogonal decomposition by total eigenvalue; a representation by fractional multiple Wiener integrals is not established.

Fractional calculus enters in two ways. The Rodrigues operation shifts the order of an individual special-function branch. We justify it on a domain that includes the negative orders in the left branch of the first eigenfunction, and show why equal branch increments fail to preserve the deformed order relation. Spectral powers and Caputo time evolution instead retain the eigenfunctions and change their decay factors [3, 22, 2]. For the one-coordinate dynamics we prove convergence to the Gaussian system at one parameter endpoint and to radial three-dimensional Ornstein-Uhlenbeck dynamics at the other. The fractional evolution operators converge in norm after the changing Hilbert spaces have been identified, uniformly on compact time intervals separated from zero.

The application observes whether each of several independent equilibrium coordinates lies on the positive half-line and multiplies the centred observations. The interface values and derivatives give its spectral coefficients without integrating the eigenfunctions over the half-line. The eigenvalues of the product operator then give the time evolution, and the omitted coefficient mass bounds the truncation error. This is the computational purpose of the example.

We compare a shared inverse stable clock with independent copies of that clock. Such clocks describe Caputo time-fractional evolution [15, Sections 2–3], within the multivariate framework of Beghin, Macci and Ricciuti [4, Sections 1, 4]. Related spectral correlation formulas are given

in [16, 19]. With equilibrium initial data, the two models have identical one-coordinate process laws and identical joint laws at each fixed time, but their collective correlations decay as $t^{-\tau}$ and $t^{-m\tau}$ for $m \geq 2$ and $0 < \tau < 1$. This rate distinction comes from the clocks. The interface expansion makes the correlation decay curves and their truncation bounds computable for the present non-Gaussian family.

Section 2 constructs the Gaussian interface basis, and Section 3 passes to the weighted probability law and product chaos. Section 4 examines Rodrigues shifts. Section 5 gives the fractional evolution and endpoint limits. Section 6 carries out the threshold computation.

2 The Gaussian interface basis

2.1 Special-function conventions and the correction to Part I

All L^2 spaces are complex Hilbert spaces, with inner products linear in the first argument; the displayed eigenfunctions are real. Write $\mathbb{N}_0 = \{0, 1, \dots\}$ and

$$w_y(x) = (2\pi y)^{-1/2} e^{-x^2/(2y)}, \quad \gamma_y(dx) = w_y(x) dx, \quad \gamma = \gamma_1.$$

We use the standard parabolic-cylinder function D_a . For $a, z \in \mathbb{C}$, it is given by (see [1, formula 19.12.1])

$$D_a(z) = \sqrt{\pi} 2^{a/2} e^{-z^2/4} \left[\frac{{}_1F_1\left(-\frac{a}{2}; \frac{1}{2}; \frac{z^2}{2}\right)}{\Gamma((1-a)/2)} - \frac{\sqrt{2}z}{\Gamma(-a/2)} {}_1F_1\left(\frac{1-a}{2}; \frac{3}{2}; \frac{z^2}{2}\right) \right].$$

Here ${}_1F_1$ is Kummer's confluent hypergeometric function of the first kind,

$${}_1F_1(b; c; z) = \sum_{n=0}^{\infty} \frac{(b)_n}{(c)_n} \frac{z^n}{n!}, \quad c \notin \{0, -1, -2, \dots\},$$

where $(b)_0 = 1$ and $(b)_n = b(b+1) \cdots (b+n-1)$ for $n \geq 1$. The reciprocal Gamma factors are understood by their entire continuation.

The power normalisation of Part I [7, Eq. (3.8)] is

$$H_a(x, y) = y^{a/2} e^{x^2/(4y)} D_a(x/\sqrt{y}), \quad a \in \mathbb{C}, \quad x \in \mathbb{R}, \quad y > 0.$$

The exponential factor gives polynomial growth on the positive half-line. At $n \in \mathbb{N}_0$,

$$H_n(x, y) = y^{n/2} \text{He}_n(x/\sqrt{y}), \quad \text{He}_n(x) = (-1)^n e^{x^2/2} \frac{d^n}{dx^n} e^{-x^2/2},$$

so $\int \text{He}_n \text{He}_m d\gamma = n! \delta_{nm}$. This is the probabilists' normalisation [17, Eq. 12.7.2 and Tables 18.3.1, 18.5.1].

For reference, the special-function facts used below are

$$\begin{aligned} D_a''(z) + (a + 1/2 - z^2/4) D_a(z) &= 0, \\ D_a(z) &= z^a e^{-z^2/4} (1 + O(z^{-2})) \quad (z \rightarrow +\infty), \end{aligned} \tag{1}$$

and, for $\text{Re } a < 0$,

$$D_a(z) = \frac{e^{-z^2/4}}{\Gamma(-a)} \int_0^\infty t^{-a-1} e^{-t^2/2-zt} dt. \tag{2}$$

The integral is not used as an ordinary integral at nonnegative orders. Primes on D_a differentiate its argument. Its origin data are

$$D_a(0) = \frac{2^{a/2}\sqrt{\pi}}{\Gamma((1-a)/2)}, \quad D'_a(0) = -\frac{2^{(a+1)/2}\sqrt{\pi}}{\Gamma(-a/2)}. \quad (3)$$

Here and below $1/\Gamma$ is the entire reciprocal-Gamma function, including at Gamma poles. These formulas are in [17, Sections 12.2, 12.5, 12.9]. The equation and recurrences imply

$$(\partial_y + \tfrac{1}{2}\partial_{xx})H_a = 0, \quad \partial_x H_a = aH_{a-1}, \quad -y\partial_{xx}H_a + x\partial_x H_a = aH_a.$$

Proposition 2.1 (Full-line integrability). *For real a and $y > 0$, $H_a(\cdot, y) \in L^2(\gamma_y)$ if and only if $a \in \mathbb{N}_0$.*

Proof. The positive tail is polynomial by (1). For real $a \notin \mathbb{N}_0$, the connection and asymptotic formulas [17, Sections 12.2, 12.9] give

$$H_a(x, y) = \frac{\sqrt{2\pi} y^{a+1/2}}{\Gamma(-a)} e^{x^2/(2y)} |x|^{-a-1} (1 + O(x^{-2})) \quad (x \rightarrow -\infty).$$

The coefficient is nonzero, and its square is not integrable against γ_y . At nonnegative integers the polynomial identity applies instead. $\square$

Consequently $dH_a(W_t, t) = aH_{a-1}(W_t, t)dW_t$, for a standard Brownian motion W and $t > 0$, is a local Itô identity and is not an identity of square-integrable martingales at noninteger real a . This alone does not exclude an L^1 martingale interpretation. Also, for nonzero deterministic real $g \in L^2([0, T])$ and $\xi = \int_0^T g(s)dW_s$, the variance argument in $H_a(\xi, y)$ is $y = \|g\|_2^2$, not $\|g\|_2$. These facts correct the variance in Definition 5 and the finite-second-moment interpretation of Theorem 7 of Part I. At integer $n \geq 1$ the unnormalised covariance is $n!s^n$ for $0 < s \leq t$, by the Hermite martingale property.

2.2 Transport, operator choice and matching

Fix $0 < \mu \leq 1$ and $y > 0$. We seek a full-line eigenfunction $\Psi_\alpha^{(\mu, y)}$ with real eigenvalue α , constructed by joining two admissible half-line branches. On the right we use $H_\alpha(x, y)$; on the left we reflect $H_{\beta_\mu(\alpha)}(u, \mu y)$, initially defined for $u > 0$. The left order $\beta_\mu(\alpha)$ will be determined by requiring both branches to satisfy the same eigenvalue equation.

The branches are initially square-integrable with respect to different Gaussian measures. To place the left branch in $L^2((-\infty, 0), \gamma_y)$, define

$$\rho_{\mu, y}(u) = \left(\frac{w_{\mu y}(u)}{w_y(u)} \right)^{1/2} = \mu^{-1/4} \exp \left[-\frac{1-\mu}{4\mu y} u^2 \right].$$

Multiplication by this factor preserves the weighted norm because $\rho_{\mu, y}^2 w_y = w_{\mu y}$. Together with reflection, it gives a unitary map between the two half-line spaces. The transported left branch is therefore

$$\rho_{\mu, y}(-x) H_{\beta_\mu(\alpha)}(-x, \mu y), \quad x < 0.$$

This identifies the Hilbert spaces. Choosing a common differential operator is a further step.

The two original branch equations have coefficients $-y$ and $-\mu y$ in front of the second derivative. Transport alone therefore does not specify a common differential expression.

We choose the common differential part $-y\partial_{xx} + x\partial_x$ on both half-lines. On the negative half-line $x < 0$, we also allow a term that multiplies the unknown function by $c_{\mu, y}x^2$ (the quadratic

potential), with $c_{\mu,y}$ to be determined. It acts only on $x < 0$, leaving the right-hand equation unchanged. Our full-line eigenvalue equation is

$$\left(-y\partial_{xx} + x\partial_x + c_{\mu,y}x^2\mathbf{1}_{\{x<0\}}\right)\Psi_\alpha^{(\mu,y)} = \alpha\Psi_\alpha^{(\mu,y)}.$$

The reason for adding this term is that differentiating the density factor in the transported left branch produces a term proportional to x^2 . We choose $c_{\mu,y}$ to compensate for it, as the following calculation shows. We add no further constant term on $x < 0$.

To determine the coefficient and the left order, write $u = -x > 0$. Reflection preserves the form of the differential expression, giving $-y\partial_{uu} + u\partial_u$. Using

$$\frac{\rho'_{\mu,y}}{\rho_{\mu,y}} = -\frac{1-\mu}{2\mu y}u,$$

the product rule shows that conjugation by $\rho_{\mu,y}$ changes the first-derivative coefficient to u/μ and introduces multiplication by

$$\frac{1-\mu}{2\mu} - \frac{1-\mu^2}{4\mu^2 y}u^2.$$

We therefore choose

$$c_{\mu,y} = \frac{1-\mu^2}{4\mu^2 y}$$

to cancel the quadratic term. For $0 < \mu < 1$, the added potential is positive on $x < 0$ and grows as x^2 ; in the operator's quadratic form it penalises mass far into the negative half-line. At $\mu = 1$ it vanishes. Applying the resulting identity directly to the left branch gives

$$\begin{aligned} & (-y\partial_{uu} + u\partial_u + c_{\mu,y}u^2)[\rho_{\mu,y}(u)H_{\beta_\mu(\alpha)}(u, \mu y)] \\ &= \rho_{\mu,y}(u) \left[\frac{1}{\mu}(-\mu y\partial_{uu} + u\partial_u) + \frac{1-\mu}{2\mu} \right] H_{\beta_\mu(\alpha)}(u, \mu y) \\ &= \frac{\beta_\mu(\alpha) + (1-\mu)/2}{\mu} \rho_{\mu,y}(u) H_{\beta_\mu(\alpha)}(u, \mu y). \end{aligned}$$

Thus the left branch has eigenvalue $[\beta_\mu(\alpha) + (1-\mu)/2]/\mu$. Equating it to the right-hand eigenvalue α determines

$$\beta_\mu(\alpha) = \mu\alpha - \frac{1-\mu}{2}. \quad (4)$$

The scaling and constant shift arise from this operator choice; unitary transport of the original branch operator alone would preserve its eigenvalue.

The order relation ensures that both branches satisfy the same differential equation away from zero. We must still determine whether their values and derivatives can be matched at the origin.

Substituting the definition of H_α , the density factor combines with the left Gaussian exponential to give the common factor $e^{x^2/(4y)}$.

Let $A_{\alpha,y}$ and $B_{\alpha,y}$ denote the real amplitudes of the right and left branches, respectively, to be chosen not both zero; their dependence on the fixed parameter μ is implicit. Absorbing the remaining constant factors into these amplitudes, we obtain

$$\Psi_\alpha^{(\mu,y)}(x) = e^{x^2/(4y)} \begin{cases} A_{\alpha,y} D_\alpha(x/\sqrt{y}), & x \geq 0, \\ B_{\alpha,y} D_{\beta_\mu(\alpha)}(-x/\sqrt{\mu y}), & x < 0. \end{cases} \quad (5)$$

We impose continuity of the function and its first derivative at zero. These conditions ensure that differentiating the joined function produces no singular terms at the interface, consistent with the prescribed differential equation. The common exponential has value one and derivative

zero there. Reflection contributes a minus sign to the left derivative, while the spatial scale contributes $\mu^{-1/2}$. The matching conditions are therefore

$$\begin{pmatrix} D_\alpha(0) & -D_{\beta_\mu(\alpha)}(0) \\ D'_\alpha(0) & \mu^{-1/2}D'_{\beta_\mu(\alpha)}(0) \end{pmatrix} \begin{pmatrix} A_{\alpha,y} \\ B_{\alpha,y} \end{pmatrix} = 0. \quad (6)$$

A nonzero matching pair exists precisely when this matrix is singular. Substituting the origin values (3) gives the equivalent condition $\Delta_\mu(\alpha) = 0$, where

$$\Delta_\mu(\alpha) = \frac{\sqrt{\mu}}{\Gamma(-\alpha/2)\Gamma((1-\beta_\mu(\alpha))/2)} + \frac{1}{\Gamma((1-\alpha)/2)\Gamma(-\beta_\mu(\alpha)/2)}. \quad (7)$$

Thus the affine order relation makes the branch equations agree, while the matching condition selects the admissible values of α . This condition is independent of y .

Neither column of the matching matrix vanishes: a nontrivial solution of the parabolic-cylinder equation (1) cannot have both value and derivative zero at the origin. At a root of Δ_μ , the matrix consequently has rank one, and the amplitudes are unique up to a common scalar. Both amplitudes are nonzero.

When the interface values are nonzero, one may take

$$(A_{\alpha,y}, B_{\alpha,y}) = (D_{\beta_\mu(\alpha)}(0), D_\alpha(0)).$$

At a matching root, if either interface value vanishes, both vanish. In this case one instead takes

$$(A_{\alpha,y}, B_{\alpha,y}) = (-\mu^{-1/2}D'_{\beta_\mu(\alpha)}(0), D'_\alpha(0)).$$

Retaining this case is necessary to recover the classical system. Indeed, at $\mu = 1$,

$$\Delta_1(\alpha) = \frac{2}{\Gamma(-\alpha/2)\Gamma((1-\alpha)/2)},$$

whose real zeros are exactly $\mathbb{N}_0$. Dividing the matching conditions by the interface value would discard all odd Hermite eigenfunctions.

The transported left branch and the order relation above correct the corresponding construction in Definition 4 of Part I. The following subsection establishes the self-adjoint realisation and completeness directly; the full-line completeness assertion in Theorem 9 of Part I is not used.

2.3 Closed form, domain and spectrum

The matching equation gives candidate eigenfunctions. To use them as an expansion basis, we must also specify a self-adjoint operator and prove completeness. The quadratic confinement supplies the compactness needed for this step.

Let $W^{1,2}(\gamma_y)$ consist of locally absolutely continuous f with $f, f' \in L^2(\gamma_y)$. The elementary Gaussian estimate

$$\|xf\|_{L^2(\gamma_y)}^2 \leq 2y\|f\|_{L^2(\gamma_y)}^2 + 4y^2\|f'\|_{L^2(\gamma_y)}^2$$

follows first for C_c^∞ functions from

$$\int x^2|f|^2 d\gamma_y = y\|f\|_2^2 + 2y \operatorname{Re} \int xf\overline{f'} d\gamma_y$$

and Cauchy–Schwarz, then by density. In particular the form

$$a_{\mu,y}[f, g] = y \int f'\overline{g'} d\gamma_y + c_{\mu,y} \int_{-\infty}^0 x^2 f\overline{g} d\gamma_y, \quad \operatorname{Dom}(a_{\mu,y}) = W^{1,2}(\gamma_y), \quad (8)$$

is closed and nonnegative: its form norm is equivalent to the displayed Sobolev norm, with constants depending on μ, y .

Theorem 2.2 (Interface realisation and spectral basis). *The self-adjoint operator associated with (8) is*

$$A_{\mu,y}f = -yf'' + xf' + c_{\mu,y}x^2\mathbf{1}_{\{x<0\}}f. \quad (9)$$

Its domain consists of $f \in W^{1,2}(\gamma_y)$ that belong to H^2 on every bounded half-interval, have $A_{\mu,y}f \in L^2(\gamma_y)$ and satisfy $f'(0-) = f'(0+)$. Value continuity already follows from $W^{1,2}(\gamma_y)$.

The resolvent is compact and the eigenvalues form a simple sequence

$$0 \leq \alpha_0(\mu) < \alpha_1(\mu) < \cdots \rightarrow \infty.$$

A real α is an eigenvalue exactly when (7) vanishes. For $k \in \mathbb{N}_0$, define the normalised matched eigenfunction from (5) by

$$e_k^{(\mu,y)} = \frac{\Psi_{\alpha_k(\mu)}^{(\mu,y)}}{\|\Psi_{\alpha_k(\mu)}^{(\mu,y)}\|_{L^2(\gamma_y)}}.$$

These functions form a real orthonormal basis of $L^2(\gamma_y)$, with the sign of the first eigenfunction chosen so that $e_0^{(\mu,y)} > 0$.

Proof. We first identify the operator domain from the form. Removing the Gaussian weight then gives a confining oscillator, whose compact resolvent and half-line solutions establish the spectral assertions.

The representation theorem for closed forms [14, Chapter VI, Section 2] gives the operator. Testing against functions supported on either side of zero gives (9) in distributions and local H^2 regularity. Integration by parts with a compactly supported smooth test function g leaves the interface term

$$yw_y(0)\{f'(0-) - f'(0+)\}\overline{g(0)}.$$

It vanishes for all g exactly under derivative matching. Conversely these domain conditions imply the form identity for compact tests; form-core density extends it to every $g \in W^{1,2}(\gamma_y)$. No condition at infinity is inserted in this argument.

For compactness and tail selection it is useful to remove the Gaussian weight. The unitary map $J_y f = w_y^{1/2} f$ gives

$$J_y A_{\mu,y} J_y^{-1} = -y\partial_{xx} - \frac{1}{2} + \begin{cases} x^2/(4\mu^2 y), & x < 0, \\ x^2/(4y), & x \geq 0. \end{cases}$$

Its form domain consists of the functions $J_y f \in H^1(\mathbb{R})$ such that $xJ_y f \in L^2(\mathbb{R})$. Indeed,

$$\begin{aligned} y\|f'\|_{L^2(\gamma_y)}^2 &= y\|(J_y f)'\|_{L^2(dx)}^2 + \frac{1}{4y}\|xJ_y f\|_{L^2(dx)}^2 \\ &\quad - \frac{1}{2}\|J_y f\|_{L^2(dx)}^2, \end{aligned}$$

first on a core and then by closure. A bounded sequence in this form norm is relatively compact locally by Rellich's theorem, while

$$\int_{|x|>R} |(J_y f)(x)|^2 dx \leq R^{-2}\|xJ_y f\|_{L^2(dx)}^2$$

controls its tails uniformly. This proves compactness.

On each half-line, the unique square-integrable Schrödinger solution is the decaying parabolic-cylinder branch, of order α on the right and $\beta_\mu(\alpha)$ on the left. Conjugation preserves value and derivative matching because $w'_y(0) = 0$. Thus (6) characterises every eigenvalue and constructs every eigenfunction, including eigenfunctions vanishing at the interface. Its rank-one nullspace proves simplicity. The first eigenfunction can be made nonnegative by minimising the form

after replacing a minimiser by its modulus. A zero of this nonnegative C^1 solution would have zero derivative, contradicting uniqueness of the ordinary differential equation. It is therefore strictly positive. Compact self-adjoint spectral theory now gives completeness. Finally $x = \sqrt{y}z$ unitarily identifies $A_{\mu,y}$ with $A_{\mu,1}$. $\square$

Decreasing μ strengthens the potential on the negative half-line. The eigenvalues therefore rise from the Hermite levels towards the positive-half-line Dirichlet levels. The next proposition makes this comparison precise.

Proposition 2.3 (Spectral bounds and endpoint limits). *For $k \in \mathbb{N}_0$,*

$$k \leq \alpha_k(\mu) \leq 2k + 1, \quad \lim_{\mu \uparrow 1} \alpha_k(\mu) = k, \quad \lim_{\mu \downarrow 0} \alpha_k(\mu) = 2k + 1. \quad (10)$$

Each eigenvalue is real analytic on $(0, 1)$ and strictly decreasing on $(0, 1]$. In particular $0 < \alpha_0(\mu) < 1$ for $0 < \mu < 1$.

Proof. Form comparison with $A_{1,y}$ gives the lower bound by min–max. For the upper bound, extend the first $k + 1$ positive-half-line odd Hermite eigenfunctions by zero to the negative half-line. Their span lies in the form domain and has largest Rayleigh quotient $2k + 1$. At $\mu < 1$, a zero lowest eigenvalue would force a constant vanishing on the negative half-line, which is impossible for a normalised eigenfunction.

The forms have a common domain and analytic dependence on $\mu > 0$. Simplicity permits analytic perturbation [14, Chapter VII, Section 4], and the Hellmann–Feynman formula gives

$$\alpha'_k(\mu) = -\frac{1}{2\mu^3 y} \int_{-\infty}^0 x^2 |e_k^{(\mu,y)}(x)|^2 d\gamma_y < 0.$$

Form convergence at $\mu \uparrow 1$ gives the Hermite limit. For the Dirichlet half-line limit, the upper bound makes the first $k + 1$ eigenfunctions uniformly bounded in $W^{1,2}(\gamma_y)$. Along any sequence $\mu \downarrow 0$, extract a subsequence converging strongly in L^2 and weakly in $W^{1,2}$. Since $c_{\mu,y} \rightarrow \infty$, each limit vanishes on $x < 0$ and has zero trace at zero. The limits remain orthonormal. For every fixed linear combination, weak lower semicontinuity bounds its Dirichlet Rayleigh quotient by $\lim_{\mu \downarrow 0} \alpha_k(\mu)$. The positive-half-line Dirichlet eigenvalues are $2k + 1$, so min–max gives the opposite bound. This uses the limiting half-line space directly, without assuming that the limiting form is densely defined on the full line. $\square$

We henceforth write $A_\mu = A_{\mu,1}$ and $e_k^{(\mu)} = e_k^{(\mu,1)}$. At $\mu = 1$ these eigenfunctions are $\text{He}_k / \sqrt{k!}$. Eigenfunctions are C^3 at the interface: (9) implies

$$f''(0\pm) = -\frac{\alpha}{y}f(0), \quad f'''(0\pm) = \frac{1-\alpha}{y}f'(0).$$

These equalities follow from the eigenvalue equation; they are not extra domain conditions.

2.4 A Brownian interpretation of the interface eigenfunctions

The space–time functions constructed from the eigenfunctions

$$\Psi_k^{(\mu)}(x, t) = t^{\alpha_k(\mu)/2} e_k^{(\mu)}(x/\sqrt{t}), \quad t > 0,$$

give a Brownian interpretation of the unshifted operator. Homogeneity and the eigenvalue equation imply

$$\left(\partial_t + \frac{1}{2} \partial_{xx} - \frac{c_{\mu,1} x^2}{2t^2} \mathbf{1}_{\{x < 0\}} \right) \Psi_k^{(\mu)} = 0.$$

The interface regularity makes $\Psi_k^{(\mu)} \in C^{1,2}((0, \infty) \times \mathbb{R})$. For $0 < s < T$,

$$\exp \left[-\frac{c_{\mu,1}}{2} \int_s^t \frac{W_r^2}{r^2} \mathbf{1}_{\{W_r < 0\}} dr \right] \Psi_k^{(\mu)}(W_t, t), \quad s \leq t \leq T,$$

is a square-integrable martingale. Indeed, Itô's formula and the product rule cancel the drift, with no local-time term because the first derivative matches. The expected time integral of the squared stochastic integrand is bounded by

$$\|(e_k^{(\mu)})'\|_{L^2(\gamma)}^2 \int_s^T t^{\alpha_k(\mu)-1} dt < \infty,$$

and $\mathbb{E}|\Psi_k^{(\mu)}(W_t, t)|^2 = t^{\alpha_k(\mu)}$. The restriction $s > 0$ keeps the singular killing coefficient away from the initial time.

3 Weighted calculus and product spectral chaos

3.1 Why the one-coordinate law changes

Finite Gaussian tensor products cause no difficulty. If $g_1, \dots, g_m$ are nonzero orthogonal real functions in $L^2([0, T])$, then $\xi_j = \int_0^T g_j dW$ are independent Gaussians of variances $y_j = \|g_j\|_2^2$. Theorem 2.2 gives the orthonormal basis

$$\left\{ \prod_{j=1}^m e_{k_j}^{(\mu, y_j)}(\xi_j) : (k_1, \dots, k_m) \in \mathbb{N}_0^m \right\}$$

of $L^2(\sigma(\xi_1, \dots, \xi_m))$, with the usual L^2 expansion and Parseval identity. This is the standard Hilbert tensor-product construction. However, for $\mu < 1$ its first eigenfunction is nonconstant, so these bases are not compatible under the ordinary cylinder embedding $F \mapsto F \otimes 1$.

The ground-state transform resolves this obstruction in two steps. First, divide each eigenfunction by the first eigenfunction and use its square as the density of the new probability law:

$$\nu_\mu(dx) = (e_0^{(\mu)}(x))^2 \gamma(dx), \quad r_k^{(\mu)} = \frac{e_k^{(\mu)}}{e_0^{(\mu)}}.$$

The law has mass one because $e_0^{(\mu)}$ is normalised, and $r_0^{(\mu)} = 1$. Cancellation of the density factors gives

$$\int r_k^{(\mu)} r_j^{(\mu)} d\nu_\mu = \int e_k^{(\mu)} e_j^{(\mu)} d\gamma = \delta_{kj}.$$

Multiplication by the first eigenfunction,

$$G_\mu f = e_0^{(\mu)} f, \quad G_\mu : L^2(\nu_\mu) \longrightarrow L^2(\gamma),$$

is unitary, so completeness is preserved as well.

Second, subtract the lowest eigenvalue before transporting the operator:

$$N_\mu = G_\mu^{-1}(A_\mu - \alpha_0(\mu)I)G_\mu, \quad \lambda_k(\mu) = \alpha_k(\mu) - \alpha_0(\mu).$$

Its domain is $\text{Dom}(N_\mu) = \{f : e_0^{(\mu)} f \in \text{Dom}(A_\mu)\}$, and

$$N_\mu r_k^{(\mu)} = \lambda_k(\mu) r_k^{(\mu)}, \quad 0 = \lambda_0(\mu) < \lambda_1(\mu) < \dots$$

The eigenvalue is now zero on constants, as required for a conservative diffusion. The index k enumerates eigenfunctions; $\lambda_k(\mu)$ is the corresponding eigenvalue shifted by the lowest eigenvalue.

3.2 The gradient and conservative diffusion

The transformed quadratic form is the weighted integral of the squared derivative. This identifies both the gradient and its adjoint, and shows that the operator generates a diffusion under the new law.

Proposition 3.1 (Closed gradient and divergence). *The derivative on $C_c^\infty(\mathbb{R})$ is closable in $L^2(\nu_\mu)$. Its closure D_μ satisfies*

$$\text{Dom}(N_\mu^{1/2}) = \text{Dom}(D_\mu) = W^{1,2}(\nu_\mu), \quad N_\mu = D_\mu^* D_\mu,$$

where $W^{1,2}(\nu_\mu)$ consists of locally absolutely continuous f with $f, f' \in L^2(\nu_\mu)$. On compactly supported smooth functions the divergence $\delta_\mu = D_\mu^*$ is

$$\delta_\mu g = -g' + \left(x - 2 \frac{(e_0^{(\mu)})'}{e_0^{(\mu)}} \right) g. \quad (11)$$

The domain of N_μ consists of $f \in W^{1,2}(\nu_\mu)$ with locally absolutely continuous f' and

$$-f'' + \left(x - 2 \frac{(e_0^{(\mu)})'}{e_0^{(\mu)}} \right) f' \in L^2(\nu_\mu); \quad (12)$$

this expression is $N_\mu f$.

Proof. The density $m_\mu = (e_0^{(\mu)})^2 w_1$ is positive and C^1 , with $m_\mu, m_\mu^{-1}, m'_\mu/m_\mu$ bounded on compact intervals. Weighted integration by parts against compact tests proves closability: if $f_n \rightarrow 0$ and $f'_n \rightarrow g$ in $L^2(\nu_\mu)$, then $\langle g, \phi \rangle = \lim_n \langle f_n, -\phi' - (m'_\mu/m_\mu)\phi \rangle = 0$. Cutoffs followed by local mollification identify the closure with the stated weighted Sobolev space.

To identify the operator, transport the closed form of $A_\mu - \alpha_0(\mu)I$. Expanding derivatives and using the equation for the first eigenfunction gives, for compactly supported H^1 functions,

$$a_{\mu,1}[e_0^{(\mu)} f, e_0^{(\mu)} g] - \alpha_0(\mu) \langle e_0^{(\mu)} f, e_0^{(\mu)} g \rangle_{L^2(\gamma)} = \int f' \overline{g'} d\nu_\mu. \quad (13)$$

The core assertion deserves attention because the spaces have changed. Approximate $e_0^{(\mu)} f$ in the original form norm by $u_n \in C_c^\infty$. Each $u_n/e_0^{(\mu)}$ has compact support and is locally H^1 ; positivity of the density permits approximation by smooth functions in the weighted derivative norm. Identity (13) then gives the same approximation in the transported form norm. Thus the transported form is exactly the closed derivative form, and its represented operator is $D_\mu^* D_\mu$.

Weighted integration by parts gives (11). Testing the represented operator locally shows that $m_\mu f'$ is locally absolutely continuous and gives (12). Conversely the displayed conditions imply the form identity on compact tests and hence on the whole form domain. This proves the maximal differential domain. $\square$

The derivative form has the Markov contraction property, and 1 belongs to its domain with zero quadratic form value. Consequently

$$e^{-tN_\mu/2} = e^{\alpha_0(\mu)t/2} G_\mu^{-1} e^{-tA_\mu/2} G_\mu$$

is a conservative symmetric Markov semigroup, with invariant reversible law ν_μ and generator

$$-\frac{1}{2}N_\mu = \frac{1}{2}\partial_{xx} + \left(\frac{(e_0^{(\mu)})'}{e_0^{(\mu)}} - \frac{x}{2} \right) \partial_x.$$

The drift is locally Lipschitz and grows at most linearly: the cylinder asymptotics in [17, Section 12.9] give $-x/2 + O(x^{-1})$ as $x \rightarrow +\infty$ and $-x/(2\mu) + O(|x|^{-1})$ as $x \rightarrow -\infty$. The corresponding stochastic differential equation, with unit diffusion coefficient, therefore has a unique nonexplosive solution and realises this semigroup. The general form and ground-state transform principles are classical [11, Chapter 1]; the calculation above supplies the domain and factorisation for this interface model. This is a weighted gradient–divergence calculus, within the non-Gaussian differentiable-measure framework [6, 10]. At $\mu = 1$ it is Gaussian calculus. For $\mu < 1$ we do not identify its divergence with a Wiener-space Skorohod integral.

3.3 The product gradient and spectral decomposition

On $\mathbb{R}^{\mathbb{N}}$ with its product Borel sigma-algebra and law $\mathbb{P}_\mu = \nu_\mu^{\otimes \mathbb{N}}$, let X_j be the coordinate maps. For a multi-index $\mathbf{k} = (k_j)_{j \geq 1}$ with only finitely many nonzero entries define

$$R_{\mathbf{k}}^{(\mu)} = \prod_j r_{k_j}^{(\mu)}(X_j), \quad \Lambda_{\mathbf{k}}(\mu) = \sum_j \lambda_{k_j}(\mu).$$

Here $R_{\mathbf{k}}^{(\mu)}$ is a product eigenfunction and $\Lambda_{\mathbf{k}}(\mu)$ is the sum of its coordinate eigenvalues. Both expressions are finite in effect, since $r_0^{(\mu)} = 1$ and $\lambda_0(\mu) = 0$. On the finite linear span of these eigenfunctions, define the coordinate gradient by

$$(\mathbf{D}_\mu^0 R_{\mathbf{k}}^{(\mu)})_j = (r_{k_j}^{(\mu)})'(X_j) \prod_{i \neq j} r_{k_i}^{(\mu)}(X_i).$$

It takes values in $L^2(\mathbb{P}_\mu; \ell^2)$.

Theorem 3.2 (Product spectral calculus). *The products $R_{\mathbf{k}}^{(\mu)}$ form an orthonormal basis of $L^2(\mathbb{P}_\mu)$. The gradient is closable, and its closure $\mathbf{D}_\mu$ and number operator $\mathbf{N}_\mu = \mathbf{D}_\mu^* \mathbf{D}_\mu$ have domains*

$$\begin{aligned} \text{Dom}(\mathbf{D}_\mu) &= \left\{ F = \sum_{\mathbf{k}} F_{\mathbf{k}} R_{\mathbf{k}}^{(\mu)} \in L^2 : \sum_{\mathbf{k}} \Lambda_{\mathbf{k}}(\mu) |F_{\mathbf{k}}|^2 < \infty \right\}, \\ \text{Dom}(\mathbf{N}_\mu) &= \left\{ F = \sum_{\mathbf{k}} F_{\mathbf{k}} R_{\mathbf{k}}^{(\mu)} \in L^2 : \sum_{\mathbf{k}} \Lambda_{\mathbf{k}}(\mu)^2 |F_{\mathbf{k}}|^2 < \infty \right\}. \end{aligned}$$

Moreover

$$\|\mathbf{D}_\mu F\|_2^2 = \sum_{\mathbf{k}} \Lambda_{\mathbf{k}}(\mu) |F_{\mathbf{k}}|^2, \quad \mathbf{N}_\mu R_{\mathbf{k}}^{(\mu)} = \Lambda_{\mathbf{k}}(\mu) R_{\mathbf{k}}^{(\mu)}.$$

Proof. Orthonormality is independence. Finite-coordinate conditional expectations converge in L^2 to every F , and each finite-coordinate space has the tensor-product basis, proving completeness. The one-coordinate factorisation yields

$$\langle D_\mu r_a^{(\mu)}, D_\mu r_b^{(\mu)} \rangle = \lambda_b(\mu) \delta_{ab}.$$

Fubini therefore gives $\langle \mathbf{D}_\mu^0 R_{\mathbf{k}}^{(\mu)}, \mathbf{D}_\mu^0 R_{\mathbf{l}}^{(\mu)} \rangle = \Lambda_{\mathbf{k}}(\mu) \delta_{\mathbf{k}\mathbf{l}}$. If finite sums tend to zero while their gradients converge to V , each coefficient tends to zero. This identity makes V orthogonal to every gradient image; it is also in the closed span of those images, so $V = 0$. Completion of the graph norm gives the first domain. The associated diagonal form gives the second domain and operator action. $\square$

Let S_μ be the set of numerical values $\Lambda_{\mathbf{k}}(\mu)$. Each bounded interval contains only finitely many of them: a sum at most R contains at most $\lfloor R/\lambda_1(\mu) \rfloor$ positive terms, from a finite list of

one-coordinate gaps. Grouping the eigenfunctions by eigenvalue gives

$$L^2(\mathbb{P}_\mu) = \bigoplus_{\Lambda \in S_\mu} \mathcal{C}_\Lambda^{(\mu)}, \quad (14)$$

$$\mathcal{C}_\Lambda^{(\mu)} = \overline{\text{span}}\{R_{\mathbf{k}}^{(\mu)} : \Lambda_{\mathbf{k}}(\mu) = \Lambda\} = \ker(\mathbf{N}_\mu - \Lambda I).$$

We call (14) the product spectral chaos. Its components are grouped by eigenvalue, which can be noninteger; the number of active coordinates remains an integer. The closure is necessary because each positive eigenvalue has infinite multiplicity under coordinate permutations. The constant subspace is $\mathcal{C}_0^{(\mu)} = \mathbb{C}1$. At $\mu = 1$, (14) is the usual Wiener chaos on the Gaussian coordinate space. Away from that endpoint it is a spectral deformation with generally noninteger eigenvalues, canonical relative to these coordinates. Indeed, sign invariance would force $e_0^{(\mu)}$ to be even, which is incompatible with the different half-line potentials. There is therefore no claimed invariance under arbitrary orthogonal changes of Gaussian coordinates or symmetric-Fock identification.

3.4 The relation to Gaussian chaos

The one-coordinate law is non-Gaussian when $\mu < 1$. A Gaussian density would force $e_0^{(\mu)}(x) = C \exp(ax^2 + bx)$ with real a, b . Its equation on $x > 0$ gives $2a - 4a^2 = 0$ and $b - 4ab = 0$. Square-integrability excludes $a = 1/2$, leaving $a = b = 0$ and a zero lowest eigenvalue, a contradiction.

Proposition 3.3 (Singularity of the countable product law). *For $0 < \mu < 1$, $\nu_\mu^{\otimes \mathbb{N}} \perp \gamma^{\otimes \mathbb{N}}$. Moreover $\prod_{j=1}^m e_0^{(\mu)}(X_j)$ does not converge in $L^2(\gamma^{\otimes \mathbb{N}})$, where X_j is the j th coordinate map on $\mathbb{R}^{\mathbb{N}}$.*

Proof. Choose a Borel set B with $\nu_\mu(B) \neq \gamma(B)$. Under the two product laws, the strong law makes $m^{-1} \sum_{j=1}^m \mathbf{1}_B(X_j)$ converge to different constants. The disjoint full-measure events prove singularity. Independence and normalisation also give

$$\left\| \prod_{j=1}^{m+1} e_0^{(\mu)}(X_j) - \prod_{j=1}^m e_0^{(\mu)}(X_j) \right\|_2^2 = 2 - 2 \int e_0^{(\mu)} d\gamma > 0,$$

where strictness follows from nonconstancy and Cauchy-Schwarz. $\square$

Thus the finite-dimensional unitary density multipliers have no corresponding infinite product multiplier on these Gaussian coordinates. This does not exclude a separate measurable coordinate transport between the probability spaces.

4 Rodrigues calculus on the spectral branches

The product basis is now fixed. We return to the special-function branches to ask whether a Rodrigues shift can move between its eigenfunctions. There are two requirements: the shift must be defined on both branches, and the shifted branches must still satisfy the order and matching conditions. The first requirement can be met even at the negative order of the left branch of the first eigenfunction. Equal nonzero increments fail the second whenever $\mu < 1$.

To specify the domain, use $\hat{h}(\xi) = \int_{\mathbb{R}} e^{-ix\xi} h(x) dx$ and define the closed $L^2(dx)$ multiplier

$$\mathcal{D}_-^s h = \mathcal{F}^{-1}[(-i\xi)^s \hat{h}(\xi)], \quad \text{Dom}(\mathcal{D}_-^s) = \{h \in L^2 : (-i\xi)^s \hat{h} \in L^2\},$$

where $(-i\xi)^s = |\xi|^s e^{-i\pi s \text{sgn}(\xi)/2}$ for real s . For $s < 0$ the domain restriction at zero frequency is essential. For $n \in \mathbb{N}$ and $n - 1 < s < n$, this agrees, when the classical expression exists, with

the right-sided Weyl derivative

$$\frac{(-1)^n}{\Gamma(n-s)} \frac{d^n}{dx^n} \int_x^\infty (t-x)^{n-s-1} h(t) dt;$$

see [21, Chapter 2]. Using the Gaussian density w_y , define

$$\mathcal{R}_y^s f = y^s w_y^{-1} \mathcal{D}_-^s(w_y f), \quad w_y f \in \text{Dom}(\mathcal{D}_-^s).$$

This condition is on the Gaussian-multiplied function, not on f in Gaussian L^2 .

Proposition 4.1 (Rodrigues identity and domain). *For real $a > -1/2$,*

$$w_y H_a(\cdot, y) = y^a \mathcal{D}_-^a w_y. \quad (15)$$

Consequently $\mathcal{R}_y^a 1 = H_a(\cdot, y)$. If also $a + s > -1/2$, then $\mathcal{R}_y^s H_a(\cdot, y) = H_{a+s}(\cdot, y)$. Composition is exact when all initial, intermediate and final exponents exceed $-1/2$.

Proof. Since $\widehat{w}_y(\xi) = e^{-y\xi^2/2}$, its multiplier of exponent a belongs to L^2 exactly for $a > -1/2$, and then belongs to L^1 as well. For $-1/2 < a < 0$, (2) gives

$$\frac{1}{\Gamma(-a)} \int_0^\infty u^{-a-1} w_y(x+u) du = \frac{y^{-a/2}}{\sqrt{2\pi y}} e^{-x^2/(4y)} D_a(x/\sqrt{y}).$$

To identify the left-hand side with the stated multiplier, damp its kernel by $e^{-\varepsilon u}$. The multiplier becomes $(\varepsilon - i\xi)^a$; dominated convergence in the squared Fourier norm is valid because $a > -1/2$. Monotone convergence of the real integral identifies the same pointwise limit, proving (15) in this interval.

For fixed x , both sides of (15) are holomorphic in a on $\text{Re } a > -1/2$. For the inverse Fourier integral, domination on compact subsets holds also after inserting the logarithmic factors from differentiation in a ; for D_a it is standard analytic dependence on the order. The identity theorem extends the equality to this half-plane. Multiplication of the Fourier symbols, with the stated domain conditions, proves the shift and composition assertions. $\square$

By (4), every left spectral order is covered, since

$$\beta_\mu(\alpha_k(\mu)) = \mu(\alpha_k(\mu) + 1/2) - 1/2 > -1/2.$$

Negative initial orders cannot be avoided. For $\mu < 1$, $0 < \alpha_0(\mu) < 1$ and (3) show $(e_0^{(\mu)})'(0) > 0$. On the reflected positive left factor this requires $D'_{\beta_\mu(\alpha_0(\mu))}(0) < 0$. Since its order lies in $(-1/2, 1)$, the origin formula implies

$$-1/2 < \beta_\mu(\alpha_0(\mu)) < 0.$$

A common positive Rodrigues increment s in the natural variance coordinate of each special-function factor produces left order $\beta_\mu(\alpha_k(\mu)) + s$. The right increment, however, requires

$$\beta_\mu(\alpha_k(\mu) + s) = \beta_\mu(\alpha_k(\mu)) + \mu s.$$

These agree exactly when $\mu = 1$. The discrepancy cannot be absorbed into an amplitude, since distinct orders solve different half-line eigenvalue equations. This proves the obstruction to equal positive increments on every eigenfunction, including the first eigenfunction; the same algebra applies to negative increments whenever the domain conditions hold. It concerns the special-function factors with the half-density retained; it is not a differentiation formula for the whole transported branch. Increments s and μs respect the affine relation but need not respect the characteristic equation or matching amplitudes. Even at $\mu = 1$, a noninteger shift leaves the discrete spectrum. This calculation concerns the specified branch shifts. Fractional evolution of the fixed basis is defined spectrally in the next section.

5 Spectral fractional dynamics and parameter limits

5.1 Spectral evolution and inverse clocks

The interface basis supplies the eigenfunctions and their eigenvalues. Standard spectral calculus now gives fractional evolution by changing the scalar factor attached to each eigenfunction. We distinguish the spatial power σ , which changes spectral rates, from the Caputo order τ , which describes inverse-clock waiting intervals. The threshold calculation uses $\sigma = 1$ and $0 < \tau < 1$.

Fix $0 < \mu \leq 1$. Throughout this subsection, norms and inner products refer to $L^2(\nu_\mu)$. For $f \in L^2(\nu_\mu)$, let $f_k = \langle f, r_k^{(\mu)} \rangle$ denote its spectral coefficients, and let $P_0 f = f_0 1$ be its projection onto the constants. Recall that $r_0^{(\mu)} = 1$, $N_\mu r_k^{(\mu)} = \lambda_k(\mu) r_k^{(\mu)}$, and $0 = \lambda_0(\mu) < \lambda_1(\mu) < \dots$.

For $0 < \sigma \leq 1$, spectral calculus defines

$$(N_\mu/2)^\sigma f = \sum_{k \geq 0} (\lambda_k(\mu)/2)^\sigma f_k r_k^{(\mu)},$$

on the domain consisting of those $f \in L^2(\nu_\mu)$ for which

$$\sum_{k \geq 0} (\lambda_k(\mu)/2)^{2\sigma} |f_k|^2 < \infty.$$

Thus the eigenfunctions remain unchanged, while their eigenvalues are replaced by fractional powers [3, 22]. The semigroup $e^{-t(N_\mu/2)^\sigma}$ gives the mild solution of

$$\partial_t u + (N_\mu/2)^\sigma u = 0, \quad u(0) = f.$$

For f in the displayed operator domain, the solution is continuously differentiable in L^2 and continuous in the graph norm, including at $t = 0$.

The probabilistic meaning follows from stable subordination. For $0 < \sigma < 1$, let $S_t^{(\sigma)}$ be the stable subordinator normalised by $\mathbb{E} e^{-a S_t^{(\sigma)}} = e^{-t a^\sigma}$ for $a \geq 0$. It is a nondecreasing Lévy process. Applying its Laplace transform to each eigenfunction gives

$$e^{-t(N_\mu/2)^\sigma} f = \int_0^\infty e^{-s N_\mu/2} f \mathbb{P}(S_t^{(\sigma)} \in ds).$$

Consequently, the fractional semigroup is obtained by running the diffusion obtained by the ground-state transform on an independent stable clock; see Schilling, Song and Vondraček [22]. At $\sigma = 1$, the clock is deterministic. Under the unitary map G_μ , the corresponding Gaussian-space generator is $-[(A_\mu - \alpha_0(\mu)I)/2]^\sigma$, with its domain determined by the same spectral construction.

To introduce waiting intervals, take a separate order $0 < \tau < 1$ and the inverse stable clock $E_t^{(\tau)} = \inf\{s \geq 0 : S_s^{(\tau)} > t\}$. Intervals crossed by jumps of the subordinator become intervals on which its inverse is constant. The clock transform is expressed through the Mittag-Leffler function

$$E_\tau(z) = \sum_{n=0}^{\infty} \frac{z^n}{\Gamma(\tau n + 1)} : \quad \mathbb{E} e^{-a E_t^{(\tau)}} = E_\tau(-a t^\tau), \quad a \geq 0.$$

These identities and the inverse-clock interpretation are discussed by Meerschaert and Straka [15, Sections 2 and 5.8].

The corresponding time derivative is the Caputo derivative

$${}^C D_t^\tau u(t) = \frac{1}{\Gamma(1-\tau)} \int_0^t (t-s)^{-\tau} u'(s) ds,$$

whenever this Bochner integral exists. We consider the initial-value problem

$${}^C D_t^\tau u + (N_\mu/2)^\sigma u = 0, \quad u(0) = f.$$

Its spectral solution is

$$u(t) = \sum_{k \geq 0} f_k E_\tau[-t^\tau (\lambda_k(\mu)/2)^\sigma] r_k^{(\mu)}. \quad (16)$$

Pollard's complete-monotonicity theorem [20] gives $0 \leq E_\tau(-a) \leq 1$ for $a \geq 0$. Parseval and dominated convergence therefore show that this series defines a bounded continuous L^2 -valued function, with $u(0) = f$ and $\|u(t)\|_2 \leq \|f\|_2$. Its Laplace transform is

$$\int_0^\infty e^{-pt} u(t) dt = p^{\tau-1} [p^\tau I + (N_\mu/2)^\sigma]^{-1} f, \quad p > 0,$$

which characterises the mild solution uniquely. Equivalently, one runs the process with semigroup $e^{-s(N_\mu/2)^\sigma}$ on an independent inverse clock. For bounded Borel data this is an ordinary expectation, which extends by contraction to $L^2(\nu_\mu)$. This is the fractional Cauchy construction of Baeumer and Meerschaert [2, Theorem 3.1].

For domain data, the strong equation requires control near the initial time. The scaling identity $E_t^{(\tau)} \stackrel{d}{=} t^\tau E_1^{(\tau)}$ and the moment $\mathbb{E} E_1^{(\tau)} = 1/\Gamma(1+\tau)$ allow differentiation of the scalar clock transform. Applying the resulting bound to the spectral coefficients gives

$$\|u'(t)\|_2 \leq \frac{t^{\tau-1}}{\Gamma(\tau)} \|(N_\mu/2)^\sigma f\|_2, \quad f \in \text{Dom}((N_\mu/2)^\sigma).$$

The bound is integrable at zero. Hence u is absolutely continuous on bounded time intervals, the operator image $(N_\mu/2)^\sigma u(t)$ is continuous, and termwise integration yields

$$u(t) = f - \frac{1}{\Gamma(\tau)} \int_0^t (t-s)^{\tau-1} (N_\mu/2)^\sigma u(s) ds.$$

The Caputo equation consequently holds strongly for $t > 0$. At $\tau = 1$, we use $E_1(z) = e^z$, recovering ordinary evolution. In every case the constant eigenfunction is preserved.

The same spectral calculus gives bounded smoothing operators. For $\theta > 0$, the Bessel and centred Riesz potentials are

$$\begin{aligned} (I + N_\mu)^{-\theta} f &= \sum_{k \geq 0} (1 + \lambda_k(\mu))^{-\theta} f_k r_k^{(\mu)}, \\ N_\mu^{-\theta} (I - P_0) f &= \sum_{k \geq 1} \lambda_k(\mu)^{-\theta} f_k r_k^{(\mu)}. \end{aligned}$$

The positive gap makes the second operator bounded; it is defined to vanish on constants. Orders add under composition within either family. Proposition 2.3 also gives the one-coordinate Hilbert–Schmidt property for the Bessel potential when $\theta > 1/2$. The standard semigroup representation of positive powers [3, 22] and the Stinga–Torrea extension theorem [24, Theorem 1.1] apply to $N_\mu/2$; in the extension formula the constant eigenfunction is retained separately from the eigenfunctions corresponding to positive eigenvalues.

The evolution and potential formulas apply to the product operator $\mathbf{N}_\mu$ by replacing $r_k^{(\mu)}$ and $\lambda_k(\mu)$ with $R_{\mathbf{k}}^{(\mu)}$ and $\Lambda_{\mathbf{k}}(\mu)$. Positive eigenvalues of the product operator have infinite multiplicity, so the Hilbert–Schmidt assertion is specific to one coordinate. For the collective calculation, the essential fact is that a shared inverse clock acts on the sum of coordinate eigenvalues for each product eigenfunction.

5.2 Truncation and an interface coefficient identity

Let P_N project onto $r_0^{(\mu)}, \dots, r_N^{(\mu)}$. Parseval and monotonicity of each multiplier give, for $0 < \sigma, \tau \leq 1$,

$$\|(I - P_N)u(t)\|_2 \leq E_\tau[-t^\tau (\lambda_{N+1}(\mu)/2)^\sigma] \|(I - P_N)f\|_2. \quad (17)$$

The corresponding Bessel bound has multiplier $(1 + \lambda_{N+1}(\mu))^{-\theta}$. For centred f , truncating $N_\mu^{-1}f$ at eigenfunction N gives L^2 and derivative errors at most $\lambda_{N+1}(\mu)^{-1}\|f\|_2$ and $\lambda_{N+1}(\mu)^{-1/2}\|f\|_2$. These are analytical bounds; numerical certification requires a certified lower bound for the omitted gap, not just a root residual.

For a threshold observation, the coefficients can be calculated from values and derivatives at the interface. For $f = \mathbf{1}_{(0,\infty)}$ put $p_\mu = \nu_\mu((0, \infty))$. Then $f_0 = p_\mu$, and the Lagrange identity for $e_0^{(\mu)}$ and $e_k^{(\mu)}$ on the positive half-line gives

$$f_k = \frac{w_1(0)}{\lambda_k(\mu)} \left[e_0^{(\mu)}(0)(e_k^{(\mu)})'(0) - (e_0^{(\mu)})'(0)e_k^{(\mu)}(0) \right], \quad k \geq 1. \quad (18)$$

Indeed $f_k = \int_0^\infty e_0^{(\mu)} e_k^{(\mu)} d\gamma$. Multiply the equation for $e_k^{(\mu)}$ by $e_0^{(\mu)}$, subtract the equation for the first eigenfunction multiplied by $e_k^{(\mu)}$, and integrate over the positive half-line. Integration by parts gives the displayed boundary flux. Polynomial growth on this half-line removes the term at infinity. No division by the interface value occurs, so eigenfunctions vanishing at the interface are included.

Since $\|f\|_2^2 = p_\mu$, (17) becomes

$$\|u(t) - P_N u(t)\|_2 \leq E_\tau[-t^\tau(\lambda_{N+1}(\mu)/2)^\sigma] \left(p_\mu - \sum_{k=0}^N |f_k|^2 \right)^{1/2}. \quad (19)$$

Every truncation preserves mean p_μ . The full solution is a half-line probability under the appropriate clocks, but a finite sum need not lie in $[0, 1]$: at $\mu = \sigma = \tau = 1$, $N = 1$, it is $1/2 + e^{-t/2}x/\sqrt{2\pi}$. Thus (19) is an L^2 error estimate, not a pointwise probability bound.

5.3 Gaussian and radial Ornstein–Uhlenbeck endpoints

The endpoints have a direct interpretation. As $\mu \uparrow 1$, the extra potential disappears and the equilibrium law becomes Gaussian. As $\mu \downarrow 0$, confinement suppresses the negative half-line; the limiting eigenfunctions of the original operator have zero trace at the origin. The ground-state transform of this Dirichlet limit produces the radial diffusion below. Because the probability spaces vary with μ , we compare operators after the unitary maps G_μ . On $x > 0$ define

$$\nu_0(dx) = 2x^2\gamma(dx), \quad G_0 f(x) = \sqrt{2}xf(x), \quad N_0 = G_0^{-1}(A_{1,+}^D - I)G_0,$$

where $A_{1,+}^D$ is the positive-half-line Dirichlet realisation of $-\partial_{xx} + x\partial_x$. The measure ν_0 is a probability law and G_0 is unitary onto $L^2((0, \infty), \gamma)$. The expression for N_0 is $-\partial_{xx} + (x - 2/x)\partial_x$; its domain is specified by the transform, without an extra boundary condition on this singular expression. Thus $-N_0/2$ is the radial Ornstein–Uhlenbeck generator in dimension three. Let ι_+ denote zero extension and ι_+^* restriction.

Theorem 5.1 (Endpoint laws and fractional operators). *The laws ν_μ converge in total variation to γ as $\mu \uparrow 1$ and to ν_0 , extended by zero on $x < 0$, as $\mu \downarrow 0$. For fixed $0 < \sigma, \tau \leq 1$ and $0 < \varepsilon < T < \infty$, the operators*

$$G_\mu E_\tau[-t^\tau(N_\mu/2)^\sigma]G_\mu^{-1}$$

converge in norm on $L^2(\gamma)$, uniformly for $t \in [\varepsilon, T]$, to

$$\begin{cases} E_\tau[-t^\tau(A_1/2)^\sigma], & \mu \uparrow 1, \\ \iota_+ G_0 E_\tau[-t^\tau(N_0/2)^\sigma]G_0^{-1}\iota_+^*, & \mu \downarrow 0. \end{cases}$$

Proof. The argument has two parts: convergence of finitely many eigenprojections, then a uniform bound on the remaining multipliers. The compactness argument in Proposition 2.3 applies

to each fixed eigenfunction. At $\mu \uparrow 1$, testing its weak equation identifies every subsequential limit as a normalised Hermite eigenfunction of eigenvalue k . At $\mu \downarrow 0$, the limit vanishes on $x < 0$, has zero trace, and testing inside $x > 0$ identifies the Dirichlet eigenvalue $2k + 1$. Simplicity gives convergence of the rank-one orthogonal projections in operator norm, independently of the choice of signs. Positivity fixes the limits of the first eigenfunction:

$$e_0^{(\mu)} \longrightarrow 1 \quad (\mu \uparrow 1), \quad e_0^{(\mu)} \longrightarrow \sqrt{2} x \mathbf{1}_{\{x>0\}} \quad (\mu \downarrow 0)$$

in $L^2(\gamma)$. The inequality $\|a^2 - b^2\|_1 \leq \|a - b\|_2 \|a + b\|_2$ proves total-variation convergence.

The limiting Gaussian representatives in the Dirichlet half-line limit are the zero extensions of $\sqrt{2} \text{He}_{2k+1} / \sqrt{(2k+1)!}$. Thus the gaps tend to k or $2k$ at the respective endpoints. For any fixed finite set of eigenfunctions, projection convergence and continuity of the scalar Mittag-Leffler function give norm convergence uniformly on $[\varepsilon, T]$.

For the tails, (10) and $0 \leq \alpha_0(\mu) < 1$ give $\lambda_k(\mu) \geq k - 1$. For $M \geq 1$ the norm of the sum with $k > M$ is at most

$$E_\tau[-\varepsilon^\tau (M/2)^\sigma],$$

uniformly in μ and $t \in [\varepsilon, T]$. It tends to zero as $M \rightarrow \infty$, and the limiting spectra obey the same bound. Truncation followed by $M \rightarrow \infty$ proves the result. $\square$

In the Dirichlet half-line limit the limiting full-line operator annihilates data supported on the negative half-line. Its initial operator is $\iota_+ \iota_+^*$, so the assertion cannot include $t = 0$ on the full-line space. The theorem concerns one coordinate; it asserts no total-variation convergence for the singular countable product laws.

6 Computing the decay of collective threshold correlations

6.1 The observation and its spectral coefficients

We use the product chaos to compute how a collective threshold observation retains correlation with its initial value. The calculation has three ingredients: the interface formula supplies the coefficients, the eigenvalues of the product operator determine their evolution, and the omitted coefficient mass bounds the truncation error. Shared and independent inverse clocks provide two dynamics to which the same expansion applies.

Fix $0 < \mu \leq 1$, $0 < \tau < 1$ and an integer $m \geq 2$. The parameter μ determines the equilibrium law ν_μ and spatial eigenfunctions; τ determines the inverse stable waiting-time mechanism. For the probability $p_\mu = \nu_\mu((0, \infty))$, define the centred threshold and its collective product by

$$f(x) = \mathbf{1}_{(0, \infty)}(x) - p_\mu, \quad F(x_1, \dots, x_m) = \prod_{j=1}^m f(x_j).$$

The value $f(x)$ is the observed threshold indicator minus its equilibrium probability. For $m = 2$, F is positive when both coordinates are on the same side of the threshold and negative when they are on opposite sides, with magnitudes determined by p_μ . Its correlation with the initial value measures persistence of this joint fluctuation. More generally, integrating out any coordinate makes F vanish, so centring isolates the contribution involving all m coordinates. We have $0 < p_\mu < 1$ and $\|f\|_{L^2(\nu_\mu)}^2 = p_\mu(1 - p_\mu)$.

Let $f_k = \langle f, r_k^{(\mu)} \rangle_{L^2(\nu_\mu)}$. Since $f_0 = 0$, subtracting p_μ leaves the coefficients of eigenfunctions corresponding to positive eigenvalues unchanged. The boundary-flux identity (18) therefore gives

$$f_k = \frac{w_1(0)}{\lambda_k(\mu)} \left[e_0^{(\mu)}(0)(e_k^{(\mu)})'(0) - (e_0^{(\mu)})'(0)e_k^{(\mu)}(0) \right], \quad k \geq 1.$$

Once the eigenfunctions have been normalised, this identity gives every coefficient from the interface data, including those of eigenfunctions vanishing at the interface. The discontinuous datum belongs to $L^2(\nu_\mu)$ but not to the gradient domain; the spectral evolution below is consequently understood in the mild sense. Its expansion is infinite even at the Gaussian endpoint; Online Resource 1, Section S.1 gives the explicit Gaussian coefficients.

Let $X_1, \dots, X_m$ be independent diffusions with generator $-N_\mu/2$, initially distributed according to $\nu_\mu^{\otimes m}$. Independently of these diffusions, take an inverse stable clock $E_t^{(\tau)}$ and independent copies $E_t^{(\tau,j)}$, $1 \leq j \leq m$. Evaluate all coordinates at $E_t^{(\tau)}$ for the shared model and coordinate j at $E_t^{(\tau,j)}$ for the independent model. Define their initial-state covariances by

$$\begin{aligned} C_{\text{sh}}(t) &= \mathbb{E} \prod_{j=1}^m f(X_j(0))f(X_j(E_t^{(\tau)})), \\ C_{\text{ind}}(t) &= \mathbb{E} \prod_{j=1}^m f(X_j(0))f(X_j(E_t^{(\tau,j)})). \end{aligned} \tag{20}$$

Division by $[p_\mu(1-p_\mu)]^m$ gives the threshold correlation coefficients. Conditioning on the clocks shows that both vectors have law $\nu_\mu^{\otimes m}$ at every fixed time, and their full one-coordinate process laws agree. The shared clock couples their histories. These fixed-time statements do not imply stationarity of the time-changed processes.

6.2 Evolution in the product chaos

The calculation applies to any nonzero real centred $f \in L^2(\nu_\mu)$, with $F = \prod_{j=1}^m f(x_j)$ and covariances (20). Unless specified otherwise, norms below are in the relevant one-coordinate or product weighted probability space. For $\mathbf{k} \in \mathbb{N}_0^m$, the product basis of Section 3 gives

$$F_{\mathbf{k}} = \prod_{j=1}^m f_{k_j}, \quad F_{\mathbf{k}} = 0 \text{ if any } k_j = 0, \quad \sum_{\mathbf{k}} |F_{\mathbf{k}}|^2 = \|f\|_2^{2m}.$$

These are cylinder functions in the constructed countable product space. A product eigenfunction $R_{\mathbf{k}}^{(\mu)}$ has eigenvalue $\Lambda_{\mathbf{k}}(\mu) = \sum_{j=1}^m \lambda_{k_j}(\mu)$; the number of active coordinates is distinct from this eigenvalue label.

Write $u_{\text{sh}}(t, x)$ and $u_{\text{ind}}(t, x)$ for the expected terminal value of F starting from x . Conditional independence and the scalar clock transform yield

$$\begin{aligned} u_{\text{sh}}(t) &= \sum_{\mathbf{k}} F_{\mathbf{k}} E_\tau[-t^\tau \Lambda_{\mathbf{k}}(\mu)/2] R_{\mathbf{k}}^{(\mu)}, \\ u_{\text{ind}}(t) &= \sum_{\mathbf{k}} F_{\mathbf{k}} \prod_{j=1}^m E_\tau[-t^\tau \lambda_{k_j}(\mu)/2] R_{\mathbf{k}}^{(\mu)}. \end{aligned}$$

All multipliers lie in $[0, 1]$. Parseval therefore extends these formulas from bounded data to arbitrary $F \in L^2(\nu_\mu^{\otimes m})$, using its spectral coefficients $F_{\mathbf{k}}$. By the product version of (16), the shared evolution is the unique bounded continuous L^2 -valued mild solution of ${}^C D_t^\tau u = -\mathbf{N}_\mu u/2$, $u(0) = F$. The independent evolution agrees on one-coordinate data, but generally does not solve this equation on eigenfunctions with several active coordinates.

For the product datum, taking the inner product with F reduces the collective computation to

$$\begin{aligned} C_{\text{sh}}(t) &= \sum_{k_1, \dots, k_m \geq 1} \left(\prod_{j=1}^m |f_{k_j}|^2 \right) E_\tau \left[-\frac{t^\tau}{2} \sum_{j=1}^m \lambda_{k_j}(\mu) \right], \\ C_{\text{ind}}(t) &= \left[\sum_{k \geq 1} |f_k|^2 E_\tau[-t^\tau \lambda_k(\mu)/2] \right]^m. \end{aligned} \tag{21}$$

The coefficients already computed at the interface are thus sufficient for every $t \geq 0$. A shared clock acts on the sum of coordinate eigenvalues; independent clocks multiply the coordinate decay factors. This is an established inverse-clock mechanism [4, Sections 1, 4]; the construction here supplies the eigenfunctions and coefficients for the non-Gaussian interface family.

Theorem 6.1 (Decay of collective correlations). *For $0 < \mu \leq 1$, $0 < \tau < 1$, integers $m \geq 2$ and nonzero real centred $f \in L^2(\nu_\mu)$,*

$$0 < C_{\text{ind}}(t) < C_{\text{sh}}(t) \leq \|f\|_2^{2m}, \quad t > 0.$$

The finite positive long-time limits are

$$\begin{aligned} \lim_{t \rightarrow \infty} t^\tau C_{\text{sh}}(t) &= \frac{2}{\Gamma(1-\tau)} \sum_{k_1, \dots, k_m \geq 1} \frac{\prod_{j=1}^m |f_{k_j}|^2}{\sum_{j=1}^m \lambda_{k_j}(\mu)}, \\ \lim_{t \rightarrow \infty} t^{m\tau} C_{\text{ind}}(t) &= \left[\frac{2}{\Gamma(1-\tau)} \sum_{k \geq 1} \frac{|f_k|^2}{\lambda_k(\mu)} \right]^m. \end{aligned} \quad (22)$$

The proof uses the one-coordinate covariance at deterministic time. With a shared clock, we first take its m th power and then average over the clock. With independent clocks, we first average and then take the power. Convexity compares these quantities, and the spectral gap controls their long-time limits.

Proof. For deterministic operational time s , the one-coordinate covariance $c(s) = \langle e^{-sN_\mu/2} f, f \rangle = \sum_{k \geq 1} |f_k|^2 e^{-s\lambda_k(\mu)/2}$ is positive and strictly decreasing. Conditioning gives

$$C_{\text{sh}}(t) = \mathbb{E}[c(E_t^{(\tau)})^m], \quad C_{\text{ind}}(t) = [\mathbb{E}c(E_t^{(\tau)})]^m.$$

Invariance and Cauchy–Schwarz bound each absolute one-coordinate expectation by $\|f\|_2^2$, so these identities hold for the stated L^2 data. Tonelli gives (21); strict Jensen gives the comparison because the clock is nondegenerate for $t > 0$ [15, Sections 2–3].

For $a > 0$, complete monotonicity and the Laplace transform $p^{\tau-1}/(p^\tau + a)$ give

$$\lim_{t \rightarrow \infty} t^\tau E_\tau(-at^\tau) = \frac{1}{a\Gamma(1-\tau)}, \quad 0 \leq t^\tau E_\tau(-at^\tau) \leq \frac{1}{a(1-e^{-1})}. \quad (23)$$

For the limit, apply the monotone density Tauberian theorem [5, Sections 1.7.2–1.7.3] to this nonnegative nonincreasing function and its transform at zero. For the bound, monotonicity implies

$$t(1-e^{-1})E_\tau(-at^\tau) \leq \int_0^\infty e^{-s/t} E_\tau(-as^\tau) ds = \frac{t}{1+at^\tau}.$$

In (21), the positive gap bounds the relevant denominators below by $m\lambda_1(\mu)$ and $\lambda_1(\mu)$, respectively. The total coefficient mass is $\|f\|_2^{2m}$, so (23) and dominated convergence prove (22). A nonzero coefficient makes each limit strictly positive. $\square$

At $m = 1$ the two covariances agree. They also agree at $\tau = 1$, when the clock is deterministic and the exponential of the sum of coordinate eigenvalues factors. The displayed asymptotics concern fixed $\tau < 1$ and are not uniform at that endpoint.

6.3 Truncating the collective expansion

To compute (21), retain $1 \leq k_j \leq N$ in the shared sum and $1 \leq k \leq N$ in the independent one-coordinate sum, giving $C_{\text{sh},N}$ and $C_{\text{ind},N}$. All omitted terms are nonnegative. Their total

mass and smallest possible eigenvalue yield

$$\begin{aligned}
0 \leq C_{\text{sh}}(t) - C_{\text{sh},N}(t) &\leq E_\tau \left[-\frac{t^\tau}{2} (\lambda_{N+1}(\mu) + (m-1)\lambda_1(\mu)) \right] \\
&\quad \times \left[\|f\|_2^{2m} - \left(\sum_{k=1}^N |f_k|^2 \right)^m \right], \\
0 \leq C_{\text{ind}}(t) - C_{\text{ind},N}(t) &\leq m \|f\|_2^{2(m-1)} E_\tau [-t^\tau \lambda_1(\mu)/2]^{m-1} \\
&\quad \times E_\tau [-t^\tau \lambda_{N+1}(\mu)/2] \\
&\quad \times \left[\|f\|_2^2 - \sum_{k=1}^N |f_k|^2 \right].
\end{aligned} \tag{24}$$

For the first bound, an omitted multi-index has one coordinate exceeding N and all other coordinates positive. For the second, bound the omitted one-coordinate sum and use $a^m - b^m \leq ma^{m-1}(a-b)$, where a and b are the full and truncated sums; here $a \leq \|f\|_2^2 E_\tau [-t^\tau \lambda_1(\mu)/2]$. Monotonicity of the multipliers and Parseval prove both bounds and uniform convergence of the increasing approximations for $t \geq 0$.

For threshold data, insert $\|f\|_2^2 = p_\mu(1-p_\mu)$ and the interface coefficients. The long-time constants in (22) have corresponding bounds: for the shared constant, multiply its omitted coefficient mass by $2/[\Gamma(1-\tau)(\lambda_{N+1}(\mu) + (m-1)\lambda_1(\mu))]$; for the independent constant, the omitted one-coordinate sum before taking its m th power is at most

$$\frac{2}{\Gamma(1-\tau)\lambda_{N+1}(\mu)} \left[\|f\|_2^2 - \sum_{k=1}^N |f_k|^2 \right].$$

These bounds use the exact spectrum and coefficients. Floating-point evaluations estimate their right-hand sides; they do not certify uncertainty in the numerical inputs.

6.4 Numerical evaluation of threshold correlations

We evaluate the threshold series for $m = 2$, $\tau = 1/2$, at the Gaussian endpoint $\mu = 1$ and the deformed law $\mu = 1/2$. The identity $E_{1/2}(-z) = e^{z^2} \text{erfc}(z) = \text{erfcx}(z)$ for $z \geq 0$ permits stable evaluation. The interface roots, normalised eigenfunctions and boundary-flux coefficients are computed with SciPy [25]. Online Resource 1 gives the complete procedure, the independent Gaussian benchmark (Section S.1), and the truncation table and interface diagnostics (Section S.2).

Figure 1 shows the computed threshold covariances, each divided by $[p_\mu(1-p_\mu)]^2$, with $N = 32$ eigenfunctions per coordinate. The bands use (24); the Gaussian quadrature points are independent of the interface calculation. The curves illustrate the $t^{-1/2}$ and t^{-1} rates proved in Theorem 6.1.

At $t = 1$ and $\mu = 1$, the absolute covariance errors are 1.17×10^{-4} (shared) and 5.94×10^{-5} (independent), within the estimated bounds 3.56×10^{-4} and 2.36×10^{-4} , respectively. Online Resource 1, Section S.2 reports the dependence on the number of retained eigenfunctions and the numerical diagnostics.

The calculation uses the product eigenfunctions to turn a collective observation into explicit scalar decay factors and error estimates. The gradient and divergence construct the conservative number operator, but this computation needs only its self-adjoint spectral realisation. The decay-rate distinction also holds for other reversible diffusions with a spectral gap; here it accompanies a computable non-Gaussian example of the constructed product chaos.

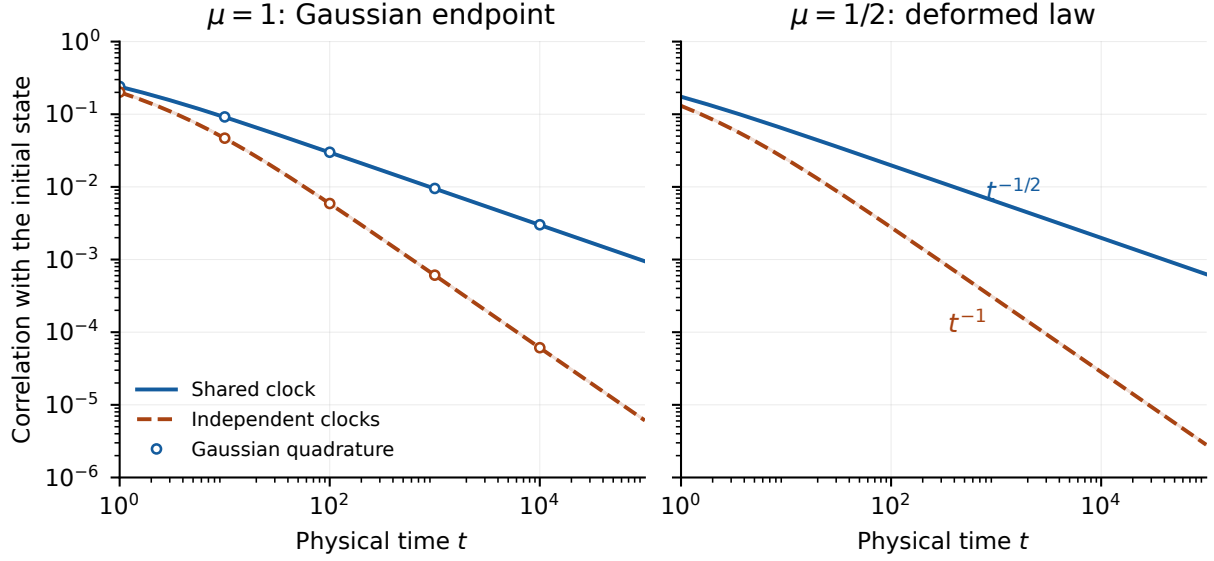


Figure 1: Collective threshold correlations for $m = 2$, $\tau = 1/2$. Solid and dashed curves use a shared clock and independent clocks, respectively. The slower common-clock decay occurs at both the Gaussian and deformed parameters. Bands are floating-point estimates of the analytical truncation bounds.

7 Concluding remarks

The corrected interface system leads to a product spectral chaos after a ground-state transform. The resulting number operator has a closed gradient–divergence factorisation and reduces to the classical Gaussian construction at $\mu = 1$. Its one-coordinate dynamics have a radial three-dimensional Ornstein–Uhlenbeck limit. The endpoint convergence also holds for the fractional evolution operators after the Hilbert spaces have been identified.

The threshold example demonstrates the computational use of this construction: interface data determine the expansion coefficients, eigenvalues of the product operator determine the decay factors, and omitted coefficient mass bounds the error. The clock dependence controls the collective decay rate. Extending the calculation to partially dependent clocks is a natural further question.

The Rodrigues analysis answers a separate structural question. Its domain includes every spectral branch, but equal nonzero increments do not preserve the deformed order relation. A fractional stochastic integral representation would therefore require a construction beyond these branch shifts and will be studied elsewhere.

Statements and Declarations

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Data and code availability. No external dataset was used. A public repository with Python code and numerical outputs for the characteristic eigenfunctions, threshold coefficients, collective clock comparison, Gaussian benchmark and truncation estimates is available at <https://github.com/colena/FWCII>.

Use of generative AI. OpenAI ChatGPT assisted with manuscript drafting and restructuring, language revision, checking mathematical arguments and references, and developing the

numerical reproduction code. The authors have verified and approved the final mathematical content, computations, citations and conclusions.

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